

1 GENERAL

$$\begin{aligned}
 CPU_t &= IC \cdot CPI \cdot T_{CLK} \\
 X \% \text{ faster than } Y &= \frac{Exec_{tY}}{Exec_{tX}} = \frac{Perf_X}{Perf_Y} \\
 CPI &= \sum_{i=1}^n (CPI_i F_i) \\
 F_i &= \frac{I_i}{IC} \\
 MIPS &= \frac{f_{CLK}}{CPI \cdot 10^6}
 \end{aligned}$$

2 AMDAHL'S LAW

$$\begin{aligned}
 Exec_t(E) &= (1 - \text{Frac}_E + \frac{\text{Frac}_E}{\text{SpUp}_E}) Exec_t(\bar{E}) \\
 \text{SpUp} &= \frac{Exec_{told}}{Exec_{tnew}} = \frac{1}{1 - \text{Frac}_E + \frac{\text{Frac}_E}{\text{SpUp}_E}}
 \end{aligned}$$

3 MIPS PIPELINING

$$\begin{aligned}
 \#_{cycles} &= IC + \#_{stalls} + 4 \\
 CPI_{as} &= \frac{m+k}{m} // \text{ m inst., k stalls} \\
 CPI_{ideal} &= \frac{1}{\#_{issues}} \\
 \text{SpUp}_{\text{Pipe}} &= \frac{\text{Depth}}{1 + \#_{stalls}}
 \end{aligned}$$

4 VLIW

$$\text{Code efficiency} = \frac{IC}{\#_{cycles} \cdot \#_{issues}}$$

5 CACHE

$$\begin{aligned}
 H_{\text{rate}} + M_{\text{rate}} &= 1 \\
 M_{\text{time}} &= H_{\text{time}} + M_{\text{pen}} \\
 AMAT_1 &= H_{\text{rate}} \cdot H_{\text{time}} + M_{\text{rate}} \cdot M_{\text{time}} \\
 &= H_{\text{time}} + M_{\text{rate}} M_{\text{pen}} \\
 AMAT_2 &= H_{\text{time}}^{L1} + M_{\text{rate}}^{L1} H_{\text{time}}^{L2} + M_{\text{rate}}^{L1L2} M_{\text{pen}}^{L1L2} \\
 MAPI &= \frac{\#_{\text{memacc}}}{IC} \\
 CPI_{\text{full}} &= CPI + \#_{\text{stalls}} + \text{Mem stalls} \\
 \text{Mem stalls}' &= MAPI M_{\text{rate}} M_{\text{pen}} \\
 \text{Mem stalls}'' &= MAPI M_{\text{rate}}^{L1} H_{\text{time}}^{L2} + \\
 &\quad + MAPI M_{\text{rate}}^{L1L2} M_{\text{pen}}^{L2}
 \end{aligned}$$

6 NETWORK

$$\begin{aligned}
 P &= \text{n. nodes} \\
 M &= \text{n. links} \\
 b &= \text{bandwidth of link} \\
 b_{\text{tot}} = Mb &= \begin{cases} b & \text{single-bus} \\ Pb & \text{ring} \\ \frac{P(P-1)}{2} b & \text{crossbar} \\ 2N(N-1)b & \text{2d mesh} \\ \frac{NP}{2} b & \text{hypercube} \end{cases} \\
 b_{\text{bis}} &= \begin{cases} b & \text{single-bus} \\ 2b & \text{ring} \\ (\frac{P}{2})^2 b & \text{crossbar} \\ Nb & \text{2d mesh} \\ 2^{N-1} b & \text{hypercube} \end{cases}
 \end{aligned}$$

7 NOTES

1. It is not possible to calculate speedup by comparing the CPIs if the clock frequencies are different.